

Skills Summary

Fitting and Interpreting Exponential Models

Use this reference while completing your worksheet or when studying.

Skill 1 — Fitting a model to two data points

Model: $y = ab^x$, where a is the value at $x = 0$ and b is the factor per unit of x .

Given $(0, a)$ and (n, f) : $ab^n = f$, so $b = \left(\frac{f}{a}\right)^{1/n}$.

Example: A colony holds 50 bacteria at $x = 0$ h and 800 at $x = 4$ h. Find b .

Step 1. The value at $x = 0$ is the coefficient, so $a = 50$; the second point then fixes b , and taking a root undoes the power.

Step 2. $50b^4 = 800 \Rightarrow b^4 = 16 \Rightarrow b = 16^{1/4} \Rightarrow b = 2$ per hour, so the model is $y = 50 \cdot 2^x$.

Try it: A tank holds 12 litres of algae at $x = 0$ h and 972 litres at $x = 4$ h. Find b .

check: 3

Skill 2 — Reading the rate from a model

Rule: in $y = ab^x$ the base b is the multiplier per unit of x .

$b > 1$: growth of $(b - 1) \times 100\%$ per unit. $b < 1$: decay of $(1 - b) \times 100\%$ per unit.

The percent is always taken from the *current* amount, so the changes multiply, never add.

Example: $A(t) = 700(1.035)^t$. State the annual percent change.

Step 1. The base is above 1, so this is growth, and the rate is how far the base sits above 1.

Step 2. $1.035 - 1 = 0.035 \Rightarrow 3.5\%$ growth per year.

Try it: $B(t) = 900(0.88)^t$. State the annual percent change.

check: 12 percent decrease per year

Skill 3 — Solving for time with logarithms

Method: to solve $ab^t = y$, divide by a , then take $\ln$ of both sides: $t = \frac{\ln(y/a)}{\ln b}$. If the exponent is t/h , multiply the result by h .

Example: Solve $30(1.15)^t = 90$ for t , to two decimal places.

Step 1. The unknown sits in the exponent, and a logarithm is the tool that brings an exponent down.

Step 2. $(1.15)^t = 3 \Rightarrow t \ln(1.15) = \ln 3 \Rightarrow t = \frac{\ln 3}{\ln 1.15} \Rightarrow t \approx 7.86$

Try it: Solve $50(1.2)^t = 200$ for t , to two decimal places.

09'2 :xəəp

(!) Watch out: $\frac{\ln 90}{\ln 30}$ is not $\ln\left(\frac{90}{30}\right)$. Divide the numbers *inside* the logarithm first, then take the logarithm.

Skill 4 — Rewriting a model with a new base

Rules: $b^t = e^{t \ln b}$, so $a b^t = a e^{kt}$ with $k = \ln b$.

And $(b^m)^t = b^{mt}$, so a model with exponent mt has yearly factor b^m .

Example: Write $400(1.05)^t$ as $400e^{kt}$. Give k to four decimal places.

Step 1. The two forms match for every t exactly when the bases match, so I need the k with $e^k = 1.05$ — that is what $\ln$ returns.

Step 2. $k = \ln(1.05) = 0.0487902 \dots \Rightarrow k \approx \mathbf{0.0488}$, so $400(1.05)^t = 400e^{0.0488t}$.

Try it: Write $250(1.12)^t$ as $250e^{kt}$. Give k to four decimal places.

8811'0 :xəəp